The Mars Lander Vacation Exercise

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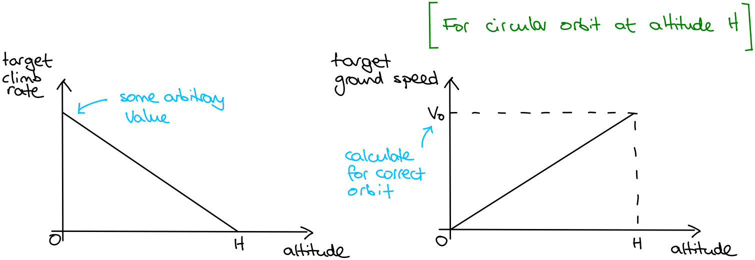
Orbital injection

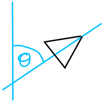


Now we have two errors to feed into our proportional controller:

e_1 = actual climb rate − target climb rate

e_2 = actual ground speed − target ground speed

We will need any-angle attitude control (see hint in lander_graphics.cpp)

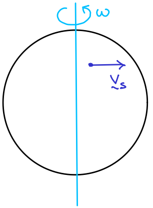


$$\theta = \tan^{-1} \frac{e_2}{e_1}$$

$$\text{throttle} \propto \sqrt{e_1^2 + e_2^2}$$

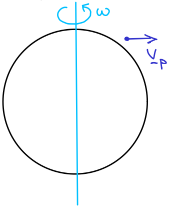
Planetary rotation and wind

① Calculate surface velocity



$\underset{\sim}{v}_s = \underset{\sim}{f}(\text{latitude})$

② Estimate atmospheric fluid velocity at position $\underset{\sim}{p}$



$\underset{\sim}{v}_p = \underset{\sim}{g}(\underset{\sim}{v}_s, \text{altitude})$

③ Add steady surface wind and gusts (optional)

$\underset{\sim}{v}_w = \underset{\sim}{v}_p +$ optional extras

④ Then, in your numerical dynamics function, tweak the drag calculation:

$$\underset{\sim}{F}_d = -\frac{1}{2}\rho C_d A |\underset{\sim}{v}_r|^2 \hat{\underset{\sim}{v}}_r$$

where $\underset{\sim}{v}_r = (\underset{\sim}{v} - \underset{\sim}{v}_w)$, the velocity of the lander relative to the atmosphere.

⑤ Must also allow for $\underset{\sim}{v}_s$ when calculating ground speed and in initial conditions.

CUED Prizewinners

. . . will be announced in the next week or two!